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DEGREE OF HUMIFICATION OF COTTON, ALFALFA AND EPHEMERS ORGANS, THEIR EFFECT ON THE CONTENT AND COMPOSITION OF SOIL ORGANIC MATTER

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ABSTRACT

The effect of plant debris on the quantity and quality of humus, the degree of humification of individual organs of cotton, alfalfa and ephemeris, their effect on the content and composition of soil organic matter was studied in the paper. The effect of plant debris of cotton and alfalfa on the humus content was studied in a small-plot field experiment. According to the results of these experiments, the balance of humus and the humification coefficient were calculated. The results obtained are of great theoretical and practical importance in solving the problem of creating a favorable humus conditions in soils and increasing their fertility; scientifically-based use of fertilizers; and the results can be used in educational process in higher educational institutions.

Keywords: Soil; organic matter; humus; humic acids; fulvic acids.

INTRODUCTION

Successful solution to the problems associated with providing the population of the country with food and agricultural raw materials is closely related to the efficient use of the farm capital assets in agriculture – the lands, to the improvement of their natural state and properties, which determine the fertility level [1-3].

Improvement of soil fertility and crop yields is closely related to the purposeful regulation of nutrient cycle in agriculture, aimed at complete supply of plants with nutrients, the main source of which is organic matter in soil [4-6]. The special role of organic matter in soil fertility is due to the fact that, being an energy source, it globally affects the complex of agronomic, physical, and biological properties of intensively used soil: it

improves the structure of arable layer, increases moisture capacity and water permeability, thereby creating air and thermal conditions favorable for plants [7-9].

The humus regime in arable soils depends on the level of agricultural technology used, including the introduction of crop rotation, the use of fertilizers, tillage, etc [10-12]. The nature of decomposition of plant debris entering the soil from cultivated crops, the degree of their influence on humus formation are also of great importance [13-15].

The land cultivation in agricultural production leads to a decrease in humus reserves in arable soils, especially in conditions of irrigated agriculture. In this regard, the development of measures to maintain a deficiency-free balance of humus in arable soils is of great importance in maintaining soil fertility.

Numerous studies carried out in Uzbekistan have established a slight decrease in humus in the upper layers of irrigated soils, in gray (typical) and hydromorphic soils with relatively high humus content; and an increase in low-humus desert soils when compared to virgin soils.

This work is devoted to the study of the effect of plant debris on the humus content in field and laboratory experiments.

The objects of research are the old-irrigated typical serozems of the gray soil zone.

Experiments were carried out on the old-irrigated typical serozem soils, in small-plots and in laboratory conditions. To conduct a small-plot

experiment, cotton plant was sown on the plot. In the fall, the mass of cotton stalks was evaluated, crushed and applied to soil, and then plowed. In the other plot, only the root mass was plowed. The control plot was the one without any crops.

In the spring of the following year, cotton was sown on all plots and humus content was periodically determined in soil: before sowing (in April), in the middle of vegetation (in July) and at the end of vegetation (in December). In 2017 cotton was cultivated on the plot applying the roots and stalks in soil. The mass of stalks applied in soil was 2.2 kg in 2017 and 1.6 kg in 2018. The mass in dry weight per plot was (5 m²), which corresponds to 3.4 t/ha.

Laboratory experiment was carried out in vessels with a capacity of 3 kg of dry soil. The soil was mixed with crushed plant debris, the mass of which, according to field estimation, corresponded to a taken sample of soil - 3 kg (Table 1).

All plant organs, except for the ephemers roots, were added in triple mass to obtain more pronounced results.

The experiment lasted 11.5 months at soil moisture content 60-80% of total moisture capacity. All laboratory and field estimations were performed according to the accepted methods described in the methodological manuals. The following indices were determined in soil samples by appropriate methods: humus content according to the I.V.Tyurin method in Nikitin's modification; humus fraction-group composition according to the I.V.Tyurin method modified by V.V.Ponomareva and T. A. Plotnikova.

Table 1. Laboratory experiment chart

No.	Options	Soil mass, kg	Mass of plant organs, g	
			Corresponding to 3 kg of soil	3 time increase
1	Soil (control)	3	-	-
2	Soil +cotton roots	3	2.2	6.6
3	Soil +cotton roots	3	3.4	10.2
4	Soil+cotton roots + stalks + leaves	3	1.2	3.6
5	Soil+roots+cotton stalks	3	4.3	12.9
6	Soil + alfalfa roots	3	6.5	19.5
7	Soil + alfalfa roots	3	5.5	16.5
8	Soil +ephemers roots	3	5.5	No increase

Aims, Research Objectives and Major Options

The aim of research is to study the effect of plant debris on the quantity and quality of humus. To achieve this aim, the following tasks were solved: to study the degree of humification of individual organs of cotton, alfalfa and ephemers, their effect on the content and composition of soil organic matter.

The Scientific Novelty of the Topic

In a small-plot field experiment, the effect of plant debris of cotton and alfalfa on the humus content was studied. According to the results of these experiments, the balance of humus and the humification coefficient were calculated. The widespread use of research results in solving the problem of creating a favorable humus condition of soils and increasing their fertility indicates the scientific and practical value of the work performed.

LITERARY REVIEW

According to figurative expression by S.Wilde, humus is "...the spirit of soil. It is one of the soil components, which, like the philosopher's stone and a homunculus, was a mystery to the alchemists and remains so until now".

Another radical judgment is gaining ground in modern literature that "... humus, humification and humic matter must be completely eliminated because these classical concepts stem from traditional procedures for fractionating soil organic matter by solubility" (von Lützow et al., 2007).

According to Waxman, "humus is a complex aggregate of amorphous matters colored in brown or dark, obtained by plant and animal debris decomposition by microorganisms under aerobic or anaerobic conditions in soil, composts, peat bogs and water basins".

I.V.Tyurin (1937) noted that "decay processes are accompanied by opposite processes of synthesis of matters that make up the microbes cells". The matters formed as a result of microbial synthesis are characterized by a significantly higher

percentage of nitrogen compared to plant debris, which explains the relative increase in the nitrogen content in soil humus.

According to Z.A.Waxman (1937), "humus and living microorganisms are connected by numerous invisible threads, which must be evaluated in order to understand the origin and nature of humus. Microorganisms affect the humus cycle in nature in several ways: 1) they contribute to its formation from plant and animal debris; 2) they cause the constant transmutation of humus under favorable conditions and in the end completely decompose or mineralize it; 3) their own cellular matter is a direct source of humus.

In the monograph by L.N.Alexandrova (1980) it is said that the difference should be distinguished between the humus formation and the formation of humic matter as a result of humification. Humus formation is a purely soil process and it occurs under a combination of soil formation factors (climate, topography, parent rock, biota, time, anthropogenic factor), while humification can occur in any objects of the environment, and humic matters can be present in many organic materials, for example, in peat, coal, sapropel, etc.

Currently, these fundamental functions of humus are supplemented by the fourth factor – an ecological one, which together form a group of soil-ecological functions, along with agroecological and biosphere-ecological ones (Semenov, Kogut, 2015).

In O.N.Biryukova, D.S.Orlov [16] the average indices of humus condition of the main soil types widespread in the Russian Federation are given. The fraction-group composition of soil humus of all soil-natural zones of flat territories is considered. Analysis and systematization of data on soil humus condition made it possible to establish a number of general patterns described by general principles and rules of humification. It is proposed to use the indices of humus condition in the development of assessment criteria of land reserves of Russia.

The intensity of humus formation processes and the amount of humus accumulation in soil are determined by a combination of factors such as

vegetation, water, air and thermal conditions, the species composition and intensity of microorganisms, mechanical composition and physicochemical properties of soil.

The low content of humus and organic nitrogen observed in poor soils of the North increases to southward in chernozem soils, and then decreases in semi-desert and steppe soils. The quality of humus varies naturally in the zonal genetic range of soils, since the degree of moisture content in organic debris, called the humus "maturity", is closely related to the conditions of humus formation. Humus is formed under high biochemical activity in soil and contains a great amount of "mature" humic acids.

The processes of decomposition and wetting of organic waste quickly develop only at sufficient soil moisture and at temperatures that support the vital activity of soil microorganisms and normal vegetation. The temperature in northern regions (where the soil moisture is sufficient and sometimes soil is oversaturated) is a factor determining the duration of the SBA. In the arid steppes, deserts and semi-deserts of southern regions, humidity is a factor determining the SBA. The maximum duration of the SBA is observed in grassy and moderately arid steppe zones, where mature species of humus, characteristic of monosomes, are formed.

Different soil types and subtypes, the distribution of which obeys the known laws of soil geography, quite naturally differ in humus reserves, in its quantitative composition and, finally, in the condition forms of the main humic matters. The processes of humus formation change (quantitatively and qualitatively), geographically,

in accordance with the general course of soil formation and, therefore, are caused by the same reasons (Tyurin, 1937).

The presence of numerous contradictions in opinions force to repeatedly revise even the basic statements concerning the very fact of existence of humic matters, their nature, ways of formation.

Plant debris in conventional field crop rotation compensates for 50-60% of humus loss as a result of mineralization. The rest is compensated by organic fertilizers application.

RESEARCH RESULTS

The influence of plowing of aboveground and underground cotton organs on the content and composition of soil humus is considered. The processes of decomposition of plant debris quickly develop only with sufficient soil moisture and at temperatures that support the vital activity of soil microorganisms. The period during which these conditions are met is called the period of soil biological activity.

The dynamics of humus content over 2 years, when applying cotton organs in soil, shows that the greatest amount of humus is observed in the spring, then there is a decrease and the least amount is observed by the end of the year (Table 2).

Obviously, this is due to the period of the highest biological activity of soil, depending on the time of the year. The same dynamics of the humus content was noted in the second year of observations, but with lower values.

Table 2. Dynamics of humus content in plowing of cotton organs, %

Option	Depth cm	2017				2018		
		24.01	24.04	26.07	6.12	10.01	15.07	6.12
Control*	0-25	1.049	1.029	1.002	1.039	1.053	1.067	1.011
	25-45	0.564	0.560	0.567	0.787	0.702	0.688	0.612
Plowing cotton stalks and roots	0-25	0.946	1.335	1.035	1.020	1.161	1.025	0.879
	25-45	0.808	1.237	0.951	0.745	1.085	0.875	0.719
Plowing cotton roots	0-25	0.946	1.329	1.003	0.994	1.063	1.013	0.860
	25-45	0.808	1.298	0.900	0.893	1.054	0.874	0.846

*In 2016, when cotton not sown for 2 previous years, the cotton roots were selected in plowing

In general, over 2 years of cotton cultivating and applying its debris in soil, applying a moderate dose of nitrogen fertilizers, there was a slight decrease in humus in the old-irrigated typical serozem. Based on the results of these estimations, the humus balance for 2 years was calculated (Table 3). In options with cotton organs application in the first year the balance was positive, and in the second year - negative. For two years, the balance of humus was negative, the deficit was 2-3 t/ha.

The dynamics in estimation of humus composition shows that no significant changes occur during the year (Table 4). A slight decrease in humic and fulvic acids amount during the growing season compared with the initial amount can be noted. Among the fractions of humic acids, the amount of the III fraction decreases more noticeably due to clay minerals and stable sesquioxides.

Changes in humus content and composition at soil composting with plant debris. The study of the organic matter content during soil composting with separate plant organs in a laboratory experiment showed that the roots of ephemeral plants gave the greatest increase in humus (Table 5). But it should be noted here that the amount of organic additives in this option is much greater in comparison with other experimental options, i.e. according to field estimations, the mass of the ephemers roots corresponding to 3 kg of soil is 25.8 g.

In addition, not all of this mass dies off annually in natural conditions, part of it enters the process of humus formation. Among the other options, the greatest growth in humus (0.046%) was obtained by applying all cotton organs to soil (option 6) and the least growth (0.005%) was obtained when applying the roots of the same crop (option 4). A noticeable increase in humus growth was observed when cotton leaves were mixed with soil (option 2).

Knowing the values of applied organic additives and data on humus growth, we calculated the humification coefficients of plant debris. It should be noted that this data is somewhat arbitrary, since the experiment was carried out in laboratory conditions and not in a field, as should be done according to the recommendations of

K.V.Dyakonova, G.Ya.Chesnyak et al. However, we can say that the data obtained give a definite idea of the humification coefficient of certain plants debris. According to the data obtained, the highest humification coefficient for cotton leaves (20%) is greater than the highest coefficient for alfalfa roots. Of the tested plant debris, the lowest coefficient was obtained for cotton roots (12%).

Analyzing the data on humus composition, it should be noted that composting the soil with cultivated plants debris did not contribute to a change in hydrolyzed matter amount, while the application of ephemeral plants roots revealed a noticeable increase in the hydrolyzability of humus matter (Table 6).

As for the humic acids, an increase in their amount is observed when introducing plant roots and especially the roots of ephemeral plants. At the same time, there is a slight decrease in the fulvic acids content. In the option with the addition of ephemeral roots, an increase in fulvic acids occurred, but in a less amount than in humic acids. In this regard, in options with composted plant roots, especially when introducing the roots of ephemeral plants, the ratio of Cha:Cfa has expanded.

In the fractional composition of humus, a slight increase in humic acids amount of the Ist fraction in soils composted with a relatively large amount of plant debris can be noted. As for fulvic acids, an increase in their amount in soil could be seen in the option when composted with the ephemers roots and their decrease in soil in all other options.

In conclusion, it should be said that in laboratory conditions the application of plant debris into soil causes an increase in soil humus. At the same time, it should be noted that high-quality humus is formed at application of plants roots.

Thus, the application of cotton plant debris, including the stalks, facilitates the humus formation; however, the experiment conducted for two years showed a negative balance of humus formed in soil, the deficit was 0.067%. A slight decrease in organic acids in humus composition is more noticeable in the IIIrd fraction, which is associated with clay minerals and sesquioxides.

Table 3. Humus balance in arable layer for 2 years (plot area 5 m²)

Options	Humus reserves, kg (on 24.01)	2017			2018			Change in 2 years	
		Introduction of plant debris, kg	Humus reserves, kg (on 6.12)	Difference	Introduction of plant debris, kg	Humus reserves, kg (on 6.12)	Difference	kg/5m ²	t/ha
Control	17.04	—	16.88	-0.16	—	16.42	-0.46	-0.62	-1.3
Plowing cotton stalks and roots	15.37	3.2	16.57	+1.20	2.6	-2.29	-2.29	-1.09	-2.1
Plowing cotton roots	15.37	1.0	16.15	+0.78	1.0	-2.19	-2.19	-1.41	-2.8

Table 4. The fraction-group composition of soil humus when plowing the cotton organs, % of total C in soil

Periods	Options	Depth, cm	C, %	Humic acids				Fulvic acids				Cha:Cfa	Hydrolyzed matter	HM:NM	
				1	2	3	sum	1 a	1	2	3				sum
Initial (January)	control	0-25	0.610	1.97	1.31	10.82	14.10	3.12	0.98	6.89	16.87	27.86	0.51	41.96	0.72
		25-45	0.396	1.55	1.04	7.77	10.36	5.18	1.55	10.36	17.62	34.72	0.30	45.08	0.82
	B-1,2	0-25	0.550	1.00	2.55	13.27	17.82	3.64	1.09	11.65	18.55	34.91	0.51	52.73	1.12
		25-45	0.470	1.49	0.64	14.68	16.81	3.83	0.43	6.38	19.15	29.79	0.56	46.60	0.87
3 months after plowing (April)	B-1	0-25	0.834	1.52	0.64	8.25	10.41	4.32	0.51	4.92	14.36	20.31	0.51	30.72	0.44
		25-45	0.730	1.53	0.28	7.23	9.04	3.89	0.42	5.17	12.72	21.70	0.42	30.74	0.44
	B-2	0-25	0.772	1.67	0.53	8.80	11.00	4.27	0.13	6.69	12.35	23.67	0.47	34.67	0.53
		25-45	0.633	1.33	0.66	7.82	9.80	3.31	0.80	9.54	19.59	21.99	0.45	31.79	0.47
11 months after plowing (December)	control	0-25	0.604	1.41	0.25	10.01	11.67	4.00	1.48	6.43	12.41	24.32	0.48	35.99	0.56
		25-45	0.457	1.81	0.24	7.81	9.86	5.43	1.27	7.75	17.29	31.74	0.31	41.60	0.71
	B-1	0-25	0.593	1.20	0.30	9.71	11.21	3.71	1.14	4.02	12.31	21.18	0.53	32.39	0.48
		25-45	0.433	1.65	0.27	5.22	7.14	3.60	0.30	5.19	12.45	21.54	0.33	28.68	0.40
	B-2	0-25	0.618	1.75	1.01	10.33	13.09	4.21	1.18	4.27	14.55	24.21	0.54	37.30	0.60
		25-45	0.519	1.31	0.71	8.58	1.60	4.51	0.60	4.81	13.84	23.76	0.45	34.36	0.52

Cha:Cfa – Carbon ratio in humic and fulvic acids

HM:NM - hydrolyzed / nonhydrolyzed matter ratio

Table 5. Change in humus content of soil upon application plant debris

№	Option	Amount of applied plant debris per 3 kg of soil, g	Initial humus content in soil 11,5 months after, %	Humus growth, %	Humus growth due to decomposition of plant debris			Humification coefficient
1	Control	-	1.109	0.004	-	-	-	-
2	Cotton leaves	2.2	1.009	0.019	0.015	0.45	608	20
3	Cotton leaves+roots	3.4	1.109	0.026	0.025	0.75	1013	22
4	Cotton roots	1.2	1.109	0.009	0.005	0.15	365	12
5	Cotton roots+stalks	4.3	1.109	0.017	0.013	0.39	689	9
6	Cotton roots+stalks+leaves	6.5	1.109	0.050	0.046	1.38	2025	21
7	Alfalfa roots	5.5	1.109	0.037	0.033	0.99	1619	18
8	Ephemers roots	25.8	1.109	0.146	0.146	4.26	5751	16

Table 6. The fraction-group composition of humus in soil composting with plant debris, % of total C of soil

№	Options	C, %	Humic acids				1a	Fulvic acids				Cha:Cfa	Hydrolyzed matter	HM:NM
			1	2	3	Sum		1	2	3	Sum			
1	Control	0.646	2.17	0.93	8.20	11.30	3.87	0.46	5.88	14.09	24.30	0.47	35.60	0.55
2	Cotton leaves	0.654	1.04	2.83	7.60	11.47	4.17	1.19	5.22	12.82	23.40	0.49	34.87	0.54
3	Cotton leaves+roots	0.658	2.34	0.59	7.17	10.10	4.10	0.15	7.03	11.71	22.99	0.44	33.09	0.50
4	Cotton roots	0.649	1.99	1.45	8.94	12.38	4.28	1.07	5.59	11.09	22.03	0.55	34.41	0.53
5	Cotton roots+stalks	0.653	1.70	1.50	8.68	11.87	4.04	0.45	5.69	11.83	22.01	0.53	33.88	0.51
6	Cotton roots+stalks+leaves	0.672	2.52	2.41	8.80	13.73	4.14	0.14	6.07	11.79	22.14	0.62	35.87	0.56
7	Alfalfa roots	0.665	1.63	2.07	8.58	12.28	3.40	1.63	4.59	11.69	21.30	0.58	33.58	0.51
8	Ephemers roots	0.728	0.82	3.16	11.98	16.66	4.26	01.82	8.01	13.87	26.96	0.62	43.62	0.77

When composting the soil with cotton organs and alfalfa and ephemers roots, the greatest growth in humus was obtained with ephemers roots, all cotton organs and alfalfa roots. The calculated humification coefficient typical for cotton leaves is - 20%, for alfalfa roots - 15%, ephemers roots - 16%, cotton roots - 12%. On lands that suffer from cotton wilt, the cotton stalks should be ploughed in a crushed form as a source of organic matter.

Thus, the application of cotton plant debris, including the stalks, facilitates the humus formation; however, the experiment conducted for two years showed a negative balance of humus formed in soil, the deficit was 0.067%. A slight decrease in organic acids in humus composition is more noticeable in the IIIrd fraction, which is associated with clay minerals and sesquioxides.

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CONCLUSIONS

1. The dynamics of humus content studied in a field experiment conducted for 2 years when applying cotton organs in soil showed that the greatest amount of humus was observed in spring time, then there was a decrease and the least amount was observed by the end of the year.
2. In options with cotton organs application there was a positive balance in the first year, and a negative balance in the second year. The humus balance for two years was negative, the deficit was 2-3 t/ha.
3. The plowing of cotton plant debris including the stalks, facilitates humus formation, however, for two years of experiment, a negative balance of humus in soil was formed; the deficit was 0.067%.

4. When composting the soil with cotton organs, alfalfa and ephemers roots, the greatest increase in humus content was obtained by applying the ephemers roots, cotton organs and alfalfa roots.
5. Composting of soil with the cultivated plants debris did not facilitate a change in hydrolyzed matter amount, whereas with ephemers roots application, a noticeable increase in the hydrolyzability of humus matter was revealed.
6. The humification coefficient typical for cotton leaves is - 20%, for alfalfa roots - 15%, for ephemers roots - 16%, for cotton roots - 12%.

COMPETING INTERESTS

Authors have declared that no competing interests exist.

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